

Communication-Matched Warp Collectives for ML-KEM and ML-DSA on a RISC-V SIMT GPU

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Abstract—ML-KEM and ML-DSA share Keccak and number-theoretic transforms (NTTs), yet their useful parallelism has incompatible communication: Keccak repeatedly reduces and permutes a 25-word state, while an NTT exchanges coefficient pairs at power-of-two distances. We present communication-matched warp collectives that keep both states in ordinary lane registers of an RV32 RISC-V GPU. The instructions are stateless, use standard source vectors and single-destination writeback, and implement fixed Keccak neighborhoods or paired NTT butterflies. A serialized modular unit shares its routing and multiplier bank across the 16-bit ML-KEM and 32-bit ML-DSA datapaths. A 200-MHz V80 evaluation covers all six standardized parameter sets against a matched, optimized cooperative software mapping. Stage, shared NTT, and arithmetic reuse accelerate complete single-request ML-KEM by $2.314\text{--}2.365\times$ and ML-DSA by $1.910\text{--}2.150\times$; eight-request batches gain $1.914\text{--}2.126\times$. A separate five-point bank sweep selects two multipliers: it removes 87.500% of NTT-hierarchy DSPs at a 2.963% standalone-cycle and at most 0.143% complete-request penalty. The results show that GPU cryptographic hardware must be sized at request scope, not by primitive peak throughput.

Index Terms—post-quantum cryptography, ML-KEM, ML-DSA, Keccak, NTT, SIMT, RISC-V

I. INTRODUCTION

ML-KEM and ML-DSA standardize lattice-based key establishment and digital signatures [1], [2]. Their deployments must process independent requests while retaining the exact rejection sampling, encoding, and control of the standards. A programmable SIMT processor is attractive because a warp can cooperate within one primitive and different warps can execute independent requests. The challenge is data movement inside the warp.

The two shared primitives expose different graphs. An NTT level contains 128 independent butterflies, but its partner distance changes at every level. Keccak-f[1600] instead applies column reductions, a fixed word permutation, and row-neighbor logic to 25 tightly coupled 64-bit words. Generic lane shuffles can express both graphs, but software pays an instruction and dependence cost for every edge. A pointer accelerator removes more instructions by moving a complete state to private storage, at the cost of transfers, memory-system integration, and a new scheduling boundary.

Prior optimized RISC-V software and scalar extensions accelerate modular butterflies and Keccak rounds [3]–[5]; GPU software distributes Keccak over 25 threads [6]; and a patent describes a warp-level Keccak round [7]. These results do not answer three questions for an open SIMT core: how a collective with multiple logical outputs fits single-destination writeback, how ML-KEM and ML-DSA share one NTT

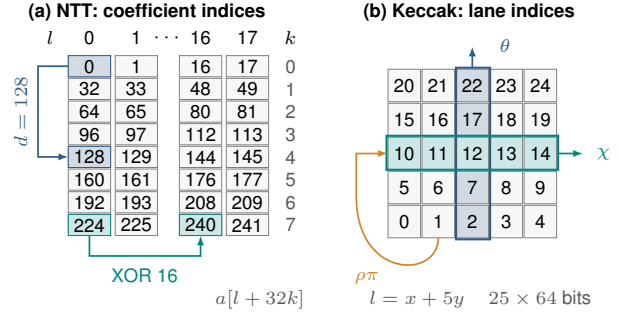


Fig. 1. Register placement determines the communication. (a) Selected NTT lanes: indices 0 and 128 pair within lane 0; 224 and 240 pair across lanes 0 and 16. (b) Keccak uses column parity (θ), a fixed permutation ($\rho\pi$, one edge shown), and row neighbors (χ).

datapath, and whether primitive throughput predicts complete-request performance.

We answer these questions in Vortex [8] with a common rule: place a primitive in lane registers, then implement only the fixed edges of its communication graph (Fig. 1). Our contributions are:

- A stateless warp-collective interface for Keccak stages and XOR-paired NTT butterflies. Each lane receives one result through normal dependency, backpressure, and writeback machinery.
- A shared K/D NTT unit whose pair routing and configurable multiplier bank support seven-layer 16-bit ML-KEM and eight-layer 32-bit ML-DSA. A five-point RTL and post-route sweep selects two multipliers.
- A six-parameter board evaluation with matched software controls, hardware ablations, and concurrency sweeps. Separate fusion and multiplier-bank studies distinguish primitive gains, complete-request benefit, and physical cost.

II. BACKGROUND AND MOTIVATION

A. Two workloads, two communication graphs

Both standards operate on degree-256 polynomials, but ML-KEM uses seven NTT layers modulo $q_K = 3329$ with 16-bit coefficients, whereas ML-DSA uses eight layers modulo $q_D = 8380417$ with 32-bit coefficients. With lane l holding $a[l + 32k]$, $0 \leq k < 8$, the first three layers exchange registers within one lane. The remaining layers exchange lane l with $l \text{ xor } d$; $d = 16, 8, 4, 2$ for ML-KEM and additionally $d = 1$ for ML-DSA. This layout gives coalesced loads and stores and keeps every intermediate coefficient in GPRs.

Both schemes also call SHA-3/SHAKE [9]. Mapping Keccak word $A[x, y]$ to lane $x + 5y$ exposes the 25-word state, but a round still needs three distinct patterns: five-lane column parity (θ), a rotate-and-gather permutation ($\rho\pi$), and three-lane row logic ($\chi\iota$). On RV32 every 64-bit word occupies two GPRs and rotations cross their boundary. A useful instruction must therefore see both halves while returning only one architectural word.

B. Why the execution boundary matters

Register residence removes inter-stage memory traffic, but it does not by itself remove communication instructions. The fair baseline is consequently a cooperative shuffle kernel with the same state placement, rather than a single-lane scalar routine. Conversely, fusing more work is not always better: it shortens the instruction stream but enlarges the unit and can serialize independent warps. A pointer engine makes the opposite trade, removing a whole permutation from the ALU path while exposing state transfers through the LSU. We evaluate all three boundaries under identical request semantics.

This reasoning also applies to the multiplier bank. A 32-lane paired butterfly needs 16 products, but peak issue rate is useful only while NTT work is on the request critical path. NTT-PIM and BP-NTT similarly choose their operation boundary from physical data placement [10], [11]. We apply that principle to the warp register file and size the arithmetic with end-to-end measurements. More generally, PacQ and SynGPU show the value of matching SIMT hardware to irregular dataflow [12], [13]; our focus is fixed cryptographic communication with exact software compatibility.

C. Relation to prior cryptographic accelerators

Scalar PQC extensions optimize a different register boundary. RISQ-V keeps polynomial state in wide scalar registers and fuses several Kyber layers, whereas Nannipieri et al. use twiddle state and packed butterfly outputs to fit a scalar instruction stream [4], [5]. Those encodings avoid a SIMT lane exchange but cannot exploit the 128 independent butterflies already present in every transform level. Our pair instruction instead treats two lanes as the natural producer of two results while preserving one destination per lane.

GPU mappings expose more parallelism but usually leave communication in software. ML-Cube maps Keccak to 25 threads and uses shuffles around machine-learning accelerators [6]; the NVIDIA collective-round description places a full 64-bit round behind a warp instruction [7]. We retain the 25-lane layout but compare three implementation boundaries: software shuffles, stage collectives, and a fused round. This comparison is necessary on RV32, where low/high results double the architectural operations and a fused unit duplicates internal work to remain stateless.

Dedicated accelerators can hide these interface costs with private SRAM and custom schedules. CHAM, NTT-PIM, BP-NTT, and ZK-Flex demonstrate the performance of co-designed polynomial storage and compute [10], [11], [14], [15]. Their boundary is appropriate when the accelerator owns

the application state. Our target retains sampling, packing, rejection control, and both standardized algorithms on one programmable core; the added units therefore operate on the same GPR state as the surrounding code. We evaluate fixed communication with software mapping and concurrency held constant, using complete requests.

D. SIMT constraints and design requirements

Vortex issues one instruction for a warp and carries a thread mask with the operand vectors. Each lane reads the same architectural register names, while the result arbiter and scoreboard treat the returned lane vector as one destination. A collective may use values from several lanes internally, but it must still retire with those ordinary semantics. Retaining a partial round or transform inside the unit would instead require a new context to be saved on a warp switch; writing two architectural destinations would require a new scoreboard and recovery rule. We therefore require one explicit transaction to consume source vectors, compute a fixed communication step, and return one value per active lane.

The active mask must contain every producer needed by that step. For Keccak, the 25 state lanes execute as one converged group and the unused lanes are padding. For an NTT at distance d , activity must be closed under $l \mapsto l \text{ xor } d$; otherwise one lane of a butterfly would read an absent partner. Distance, direction, modulus, and Keccak round index are therefore explicit instruction fields rather than hidden unit state. These rules are simple enough to assert in simulation and allow a stalled collective to carry the same warp ID, mask, and destination metadata as the existing variable-latency units.

The experiments preserve library semantics and inputs. The NTT bank sweep changes only internal scheduling, keeping the binary fixed. We select hardware using complete requests, routed core cost, and model parity.

III. DESIGN AND IMPLEMENTATION

A. Collective contract and core integration

Figure 2 places the units beside the integer ALU. An instruction consumes complete source vectors and the active mask, then returns one destination vector through the existing result arbiter. No operation contains a pointer, writes memory, or retains architectural state after writeback. Keccak requires a converged warp; NTT requires pair-closed masks. Padding lanes return zero. These constraints permit static wiring and allow different warps to interleave without saving a cryptographic context.

The decoder uses one custom opcode family and a compact function field for operation, low/high selection, NTT direction, modulus, and XOR distance. The scoreboard sees the same source and destination registers as the execution unit; no implicit register or memory dependence is introduced. A collective accepts an instruction only when its entire input vector and metadata can be captured. It holds warp ID, destination, mask, and lane results under output backpressure, so a stalled result cannot be paired with a later warp. This property is

especially important for a serialized NTT instruction, which occupies several internal beats but commits exactly once.

B. Keccak: stages, rounds, and RV32 halves

For coordinates modulo five, the datapath implements

$$\begin{aligned} C[x] &= \bigoplus_y A[x, y], \\ T[x, y] &= A[x, y] \oplus C[x-1] \oplus \text{rol}(C[x+1], 1), \\ B[y, 2x+3y] &= \text{rol}(T[x, y], \rho[x, y]), \\ A'[x, y] &= B[x, y] \oplus (\neg B[x+1, y] \wedge B[x+2, y]). \end{aligned} \quad (1)$$

The *Stage* interface exposes θ , $\rho\pi$, and $\chi\iota$. Each has low/high RV32 forms returning the selected half. Six issues form a round. The *Round* interface fuses all stages and uses a literal round index. Its low/high instructions recompute the internal full word from identical operands, avoiding hidden state and multi-destination writeback. Two issues form an RV32 round. A memory-coupled engine that reads and writes the 200-byte state provides the coarser comparison point.

Stage has two pipeline registers and accepts one operation per cycle. Round registers parity, θ , $\rho\pi$, and $\chi\iota$ in four steps and also has initiation interval one. Its larger internal state explains why it can improve an isolated permutation while producing little request-level gain. For θ and $\rho\pi$, both half instructions read both old halves because rotation carries cross the 32-bit boundary. Software therefore writes the first result to a temporary until the second issue has consumed the old pair. $\chi\iota$ reads one half and a warp-uniform round index. Round encodes the index as an immediate, enabling the 24 rounds to be expanded without a table load or branch in the permutation body.

C. NTT: one product, two lane results

For an XOR pair, pair-low lane p supplies a and twiddle ζ ; pair-high lane $h = p \text{ xor } d$ supplies b . A forward instruction computes

$$t = \text{Mont}(b\zeta), \quad (r_p, r_h) = (a + t, a - t). \quad (2)$$

One product thus serves both lanes, and each lane writes its own destination. The inverse instruction first forms a sum and difference, then multiplies the appropriate branch. Lane-local NTTMUL.K/D handles the first three layers, inverse scaling, ML-KEM base multiplication, and independent ML-DSA pointwise products.

A bank of M signed 32×32 multipliers serializes the 16 logical pair products without changing the ISA or binary. CT and ML-DSA inverse butterflies use $16/M$ beats; ML-KEM inverse and 32-lane scalar products use $32/M$. The direct latency and initiation interval are $6 + B$ and B , where B is the beat count. Operands are captured on the first handshake, lane results are assembled by index, and backpressure freezes the serializer and product pipeline. Pair routing, capture, and writeback are shared; only the constant-arithmetic reductions for q_K and q_D differ.

The shared bank preserves each library's native residue representation rather than converting coefficients at its boundary.

TABLE I
COLLECTIVE INTERFACES ON RV32. ISSUE COUNTS ARE PER KECCAK ROUND OR PER CROSS-LANE NTT LAYER; B IS THE SERIALIZED BEAT COUNT.

Operation	Cooperation	Result/lane	Issues or II
$\theta, \rho\pi, \chi\iota$	25 lanes	word half	6/round
Round	25 lanes	word half	2/round
NTTB.F.K/D	XOR pair	coefficient	$\Pi=B$
NTTMUL.K/D	lane-local	product	$\Pi=B$

For ML-KEM, operands are sign-extended from 16 bits; a shift-add Montgomery factor and the constant 20159 Barrett product produce the forward and inverse results. ML-DSA uses the full signed 32×32 product, its Montgomery inverse, and q_D in the reduction stage. Both modes therefore use the same operand capture, multipliers, beat controller, and result assembly. Modulus selection changes only the constant paths after multiplication. This organization lets NTTMUL.D serve both butterfly products and ordinary pointwise products without a second arithmetic unit.

Twiddles remain explicit lane operands. The unit contains neither a twiddle ROM nor a layer counter, so it does not assume a transform schedule or retain state between issues. Software can use the same operation for forward and inverse transforms, final scaling, ML-KEM base multiplication, and ML-DSA pointwise work. More importantly, the scheduler may suspend a warp between layers and issue another warp without saving an accelerator context.

The paired encoding addresses a scalar instruction's operand mismatch. A logical butterfly has three values (a, b, ζ) and two results, but the pair-low lane provides the twiddle and the two physical lanes independently receive r_p and r_h . This converts an awkward three-input/two-output scalar operation into an ordinary two-vector/one-vector SIMD transaction. Figure 3 shows M2 assigning successive pairs to two multipliers over eight beats, and the RV32 temporary that preserves the old Keccak state. Result assembly records their lane indices and releases the vector only after the final reduction; later warps can still execute other functional units while the issuing warp waits on the destination dependency.

D. Complete-library integration

The native libraries retain their public APIs and byte-level outputs [16], [17]. Lane zero executes scalar control, broadcasts primitive arguments, activates the warp, and restores leader execution (Fig. 4). One-shot SHAKE keeps its state in registers across absorb, permutation, and squeeze; an incremental context is stored between calls. NTT kernels load eight coefficients per lane and perform all layers without intermediate memory. ML-DSA also reuses NTTMUL.D for ordinary pointwise products. The ML-DSA-65 five-term matrix-vector sum accumulates in 64-bit software and performs one final Montgomery reduction, preserving the library representation.

Cooperation is confined to primitive kernels. All 32 lanes enter and leave a region together; Keccak activates lanes 0–24, while an NTT butterfly uses a pair-closed mask at its encoded

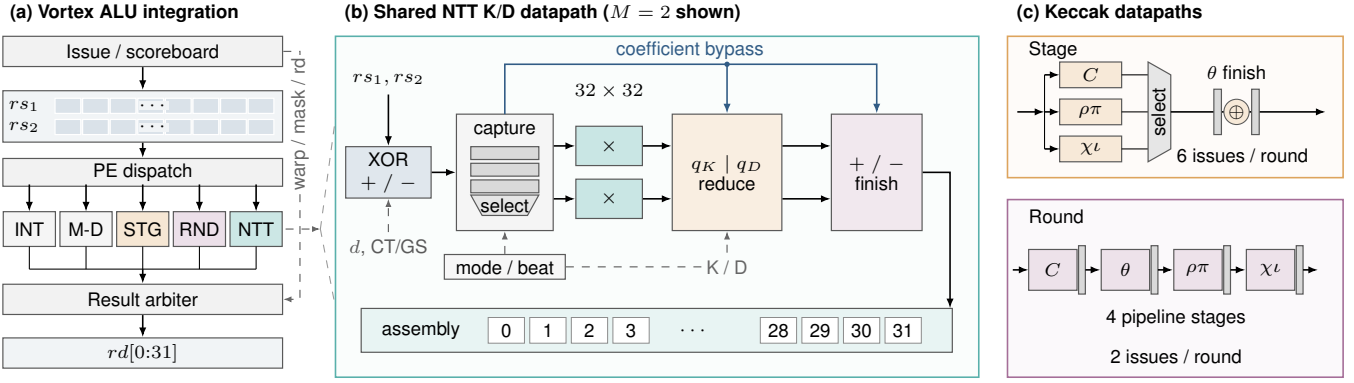


Fig. 2. Core integration and collective datapaths. (a) Dedicated PEs share operand dispatch and result arbitration with INT and multiply/divide. (b) Pair preparation includes GS pre-add/subtract. Both moduli share the multiplier bank, coefficient bypass, and result assembly. (c) Stage forms column parity C before its first register and completes θ after it; Round pipelines all four steps. Thin vertical bars denote pipeline registers.

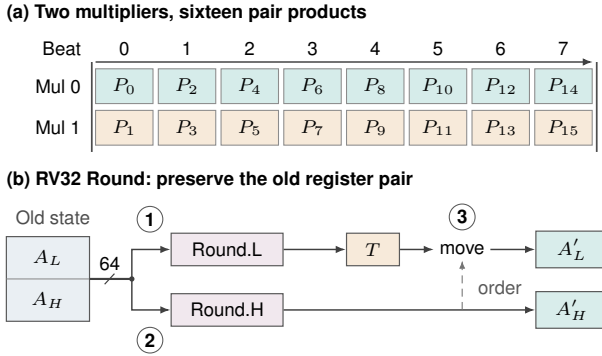


Fig. 3. Serialization and dependencies. (a) M2 assigns 16 pair products to two multipliers in eight CT or DSA-GS beats; pipeline latency is omitted. (b) Program order for an RV32 Round: both issues read the old halves, the low result remains in T , and the final move updates A_L .

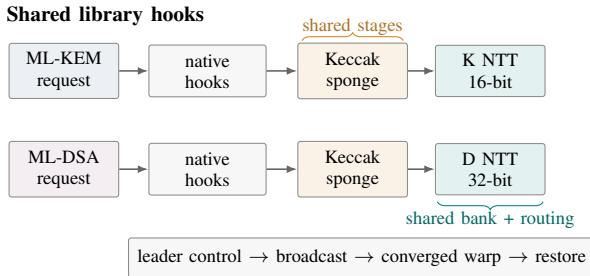


Fig. 4. Software integration. Both native libraries retain leader control and enter converged warp kernels through their existing Keccak and NTT hooks. The algorithms select a reduction mode while sharing routing and multipliers.

XOR distance. The leader controls allocation and rejection loops; warp kernels also parallelize candidate filtering, encoding, noise generation, and signing arithmetic. This keeps public interfaces unchanged and prevents a collective from observing a partially active pair. The compiler intrinsics name every source and destination, so the existing scoreboard orders a later consumer exactly as it does for an integer ALU result.

One-shot sponges and transforms keep state in GPRs between initial load and final store. Pointer Keccak instead trans-

fers the 200-byte state through the LSU, allowing independent ALU work from another warp.

Full-RAM ML-DSA reserves an 88-KiB arena per request for sets 44/65 and 160 KiB for set 87. Leader-only allocation and reverse-order release reclaim temporaries after every signing attempt. Simulation assertions check converged Keccak masks, pair-closed NTT masks, uniform round indices, backpressure, and warp metadata. Unit tests cover all pair distances and FIPS-202 rounds. Complete runs of all six parameter sets compare keys, ciphertexts or signatures, and shared secrets byte for byte with portable C.

E. Model and RTL verification

The SimX timing model mirrors every RTL bank beat, pipeline latency, and result stall. This provides a functional oracle before processor RTL runs and makes cycle disagreement observable rather than hidden by a looser simulator. The processor tests then exercise instruction decode, scoreboard hazards, masks, and writeback arbitration. XRT adds the AFU integration used by the complete applications. We require identical retired instruction counts for matched SimX/XRT binaries and keep timing gaps below the pre-defined 5% parity bound; we do not widen the bound to accommodate a new unit.

Unit testbenches cover all five XOR distances, both directions and moduli, masks, backpressure, and consecutive warp metadata. Instruction tests compare every lane with scalar references; transform boundary vectors precede complete library KATs. All 60 single/eight-request A–E cases across the six sets pass both models with identical instruction counts and at most 4.422% cycle difference. Their five board repetitions also match the XRT instruction counts. Independent post-route cores measure each hardware choice.

IV. EVALUATION

A. Methodology

The board experiments use one RV32IM Vortex core with eight 32-thread warps, 16-KiB instruction/data caches and local memory, a write-through D-cache, no L2/L3, and F/D

TABLE II
PRIMARY EXPERIMENTAL CONFIGURATION.

Processor	RV32IM, one core, F/D disabled
Warp organization	8 resident warps, 32 lanes
On-core memory	16 KiB I\$, D\$, and local memory
Workloads	ML-KEM-512/768/1024; ML-DSA-44/65/87
Timing	XRT RTL integration; SimX controls; V80 silicon
Physical flow	Alveo V80, 200 MHz, Vivado 2025.1

disabled (Table II). The shared NTT bank has two multipliers. Independent cores and the complete AVED image use Vivado 2025.1 OPT3 at 200 MHz; the board clock is read back before and after measurement. A complete request includes keypair, encapsulation, and decapsulation for KEM, or keypair, signing, and verification for DSA. We report device-side makespan, excluding host transfers, as the median of five runs after one warmup.

All arms share warp sampling, KEM encoding/noise/linear arithmetic, and DSA SHAKE extraction, pointwise accumulation, and signing arithmetic. Zeroization uses the warp with one worker and the leader otherwise. A uses software shuffles for Keccak and NTT; B adds Stage, C adds shared NTT, D adds both, and E also uses NTTMUL outside transforms. Opcode audits ensure A contains no PQC instructions and A/B contain no NTT instructions, including Montgomery conversion. Arms share the same ordered input cohort and board image. All 156 board cells (780 timed runs) pass byte-exact reference checks. XRT validates the 60 A–E cases at one/eight requests. Separate bank-sizing and fusion controls use ML-KEM-768/ML-DSA-65 and retain their archived write-back cache configuration.

B. Communication collectives improve complete requests

Figure 5 uses a single complete-mapping cohort. At one request, E accelerates ML-KEM by 2.314–2.365 $\times$ and ML-DSA by 1.910–2.150 $\times$ over A. At eight requests, the ranges are 2.004–2.126 $\times$ and 1.914–2.023 $\times$, respectively. Accelerated single-request latency is 5.912–14.757 ms for KEM and 51.264–107.181 ms for DSA.

The ablation separates each hardware contribution. Across parameter sets and one/eight requests, Stage alone gives 1.827–2.124 $\times$ for KEM and 1.466–1.744 $\times$ for DSA (A/B). With Stage fixed, shared NTT adds 1.082–1.153 $\times$ for KEM and 1.118–1.194 $\times$ for DSA (B/D). NTT therefore retains measurable request-level value after Keccak acceleration; these gains do not establish that it becomes the dominant remaining phase.

Reusing the modular unit outside transforms adds 1.040–1.146 $\times$ for DSA (D/E), with no new RTL. KEM’s corresponding range is only 0.998–1.024 $\times$; set 768 is within 0.2% of D. Arithmetic reuse is thus useful but not uniformly profitable. All headline results use E. Within each matched case, $(A/B)(B/D)(D/E) = A/E$; ratios from other mapping or input cohorts are not combined with these results.

Figure 7 separates occupancy from batch size. On eight fixed inputs, increasing workers from one to eight raises

TABLE III
INDEPENDENT RV32IM POST-ROUTE CORES AT 200 MHz. M16 VARIES KECCAK GRANULARITY; M2 IS MEASURED BOTH ALONE AND WITH STAGE.

Variant	LUT	FF	DSP	WNS (ns)
M16 base	233,759	136,451	176	+0.078
M16 + Stage	237,073	138,503	176	+0.116
M16 + Round	236,625	142,541	176	+0.052
M16 + Pointer	243,399	139,407	176	+0.080
M2 base	223,964	131,306	106	+0.048
M2 + Stage	226,852	133,368	106	+0.145

E throughput by 2.218–3.042 $\times$. With eight workers fixed, extending batches from eight to 64 changes throughput to only 0.998–1.047 $\times$ its eight-request value. Intermediate DSA batches peak at 1.119 $\times$, so throughput is not monotonic. Batches use synchronized waves. DSA signing retries vary by input; extending a batch also extends its input prefix, so its curve includes both amortization and workload variation.

C. Primitive peaks do not select the hardware

Figure 6(a) compares Round with equally unrolled Stage. At one and eight active warps, fusion improves dependent permutation throughput by 3.168 $\times$ and 3.291 $\times$, respectively. Its matched complete-KEM gains are only 1.028 $\times$ and 1.033 $\times$. In a separate instrumented control, Stage permutation time is only 4.964% of a request; sponge I/O, sampling, packing, and control limit the remaining benefit. This profile is not a phase breakdown of the six-parameter board cohort. Fixed-mapping XRT controls also find Round 1.130 $\times$ faster than Pointer at one request, but Pointer 1.374 $\times$ faster at eight. Fusion and offload therefore require request-level evaluation at the intended concurrency.

M16 is the fastest NTT bank, but M8 and M4 increase standalone NTTBF cycles by only 0.181% and 0.715%. M2 increases them by 2.963% while removing 71.074% of NTT LUTs, 46.131% of flip-flops, and 87.500% of DSPs. Across ML-KEM/ML-DSA at one/eight requests, its largest XRT makespan increase is 0.143%. M1 saves only five additional core DSPs but raises standalone cycles by 25.043% and increases NTT flip-flops due to deeper serialization. We therefore select M2, despite M16 having the highest primitive throughput.

All five standalone runs retire 59,534 instructions and pass 1,044 butterfly vectors. SimX and RTL differ by at most 0.959%. The flat request curve therefore reflects latency hiding and limited remaining NTT work, not a change in the executed instruction stream.

Every core in Table III closes without routing errors. At M16, Stage, Round, and Pointer add 1.418%, 1.226%, and 4.124% LUTs. M2 reduces whole-core LUTs and DSPs by 4.190% and 39.773% relative to M16. The directly routed M2+Stage core adds 1.289% LUTs over M2. The separate board AFU contains the optional Round and Pointer units as well; it occupies 243,073 LUTs, 181,881 FFs, and 106 DSPs and closes at 200 MHz with 0.001 ns setup slack. Board

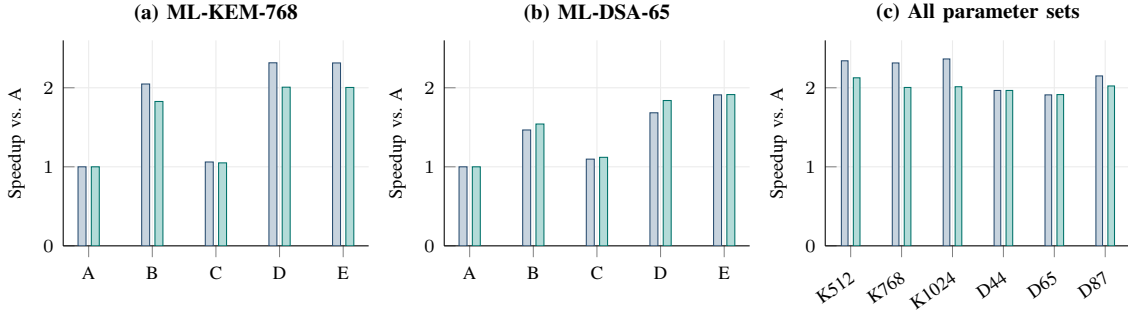


Fig. 5. 200-MHz V80 complete-request speedups over matched cooperative software (A). B enables Keccak Stage; C enables shared NTT; D enables both; E also uses modular products outside transforms. Blue/teal denote one/eight requests. (a,b) A–E ablations; (c) A/E across all six sets. Every arm uses the same complete software mapping; speedups divide five-run median makespans.

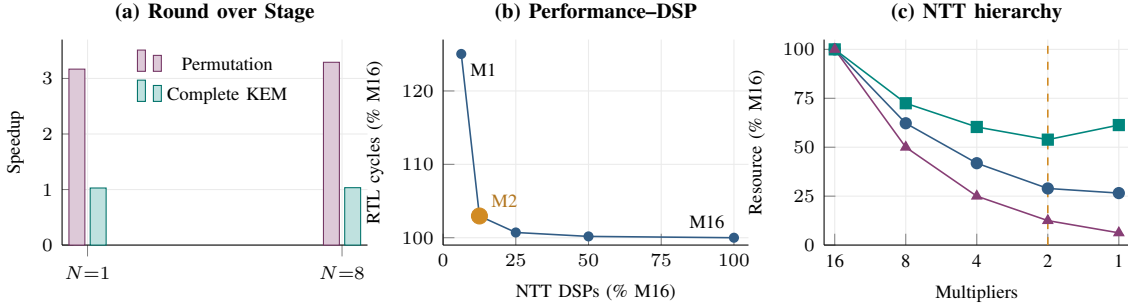


Fig. 6. Selecting granularity and multiplier count. (a) Round fusion's XRT permutation and SimX request gains. (b,c) Five banks run the same NTTBF binary; post-route NTT resources are normalized to M16. Circles, squares, and triangles denote LUTs, FFs, and DSPs in (c).

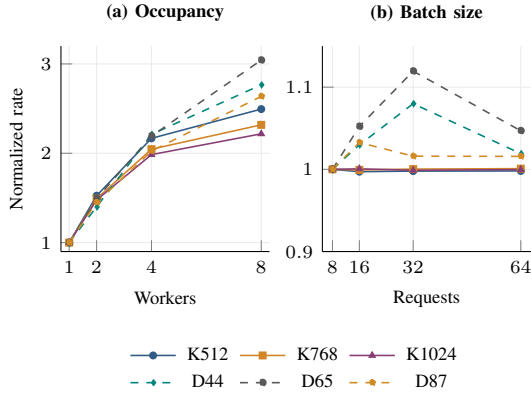


Fig. 7. 200-MHz board throughput for E. (a) Eight fixed inputs, normalized to one worker. (b) Eight workers, normalized to an eight-request batch. Curves cover all six parameter sets.

runtime ratios select software paths within this all-feature image, not different core areas.

D. Design implications and scope

Bank sizing is workload-specific: fewer warps or more transform work could move the M2 knee. The six-parameter board results validate the selected design; they do not establish an optimal bank size for every parameter and workload.

Energy requires workload activity; device timings do not measure host-to-host latency.

V. CONCLUSION

Communication-matched collectives keep Keccak and NTT state in ordinary lane registers and share one modular unit across ML-KEM and ML-DSA. At 200 MHz, all six parameter sets gain $1.910\text{--}2.365\times$ for complete single requests over matched cooperative software. Controlled ablations show incremental NTT benefit after Stage, while bank sizing and batch sweeps favor modest hardware and explicit warp concurrency. Primitive throughput alone does not predict the best complete-request design.

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